Jiawen Wang

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PROFILE

Master of Information Technology student at UTS, with graduation expected in December 2026 and a biotechnology background. Project experience in data analysis and machine learning, including comparative research, model evaluation and communicating findings.

EDUCATION

University of Technology Sydney 2025 - Present
Master of Information Technology
Selected coursework: Deep Learning, NLP Algorithms, Advanced Data Analytics, Data Visualisation and Visual Analytics.

Huazhong Agricultural University 2019 - 2023
Bachelor of Biotechnology

PROJECTS

CricSee | Cricket Analytics Research and Web Prototype

Python, pandas, scikit-learn, XGBoost, React, TypeScript, FastAPI, OpenAI API

- Built a 10-stage pipeline to evaluate remaining-player resources for win-probability prediction across **1,521 T20I matches and 360,721 events**, using pre-event features and strictly prior-date player histories.
- Compared eight feature groups and four model families using chronological splits and match-level bootstrap; found no consistent test improvement from combined player-resource features and documented calibration limitations.
- Developed a React/TypeScript and FastAPI prototype for match replay, win-probability and partial contribution analysis with OpenAI API Q&A; communicated findings to an academic supervisor through a report, presentations and video.

Image Classification and Object Detection

Course project | PyTorch, TorchVision, ResNet18, Faster R-CNN, SSD300

- Addressed unstable validation performance with an added convolutional block, BatchNorm, dropout and adjusted training settings; increased test accuracy from **75.30% to 92.43%** on the same split of a 1,670-image, 10-class dataset.
- Implemented Faster R-CNN and SSD300 training pipelines with Pascal VOC annotation parsing, invalid-box filtering, early stopping and validation-loss checkpointing to manage training and inspect localisation results.

Handwritten Mathematical Symbol Recognition

Course project | Python, scikit-learn, scikit-image

- Compared HOG, LBP and raw-pixel features with SVM and ANN classifiers on **5,000 images across 10 classes**; HOG with SVM achieved **99.3% test accuracy**, versus **96.1%** with raw pixels.
- Used a stratified 80/20 split, training-only feature scaling and confusion matrices; found shape and edge features more effective than local texture features on this dataset.

TECHNICAL SKILLS

Data & BI: Python, SQL, pandas, NumPy, Excel, Tableau, Matplotlib, Seaborn

ML & Evaluation: scikit-learn, PyTorch, XGBoost, feature engineering, chronological validation, calibration diagnostics

Development & Tools: React, TypeScript, FastAPI, OpenAI API integration, Git/GitHub, Linux/Unix, Jupyter/Colab, AI-assisted development; basic AWS exposure

Languages: English, Mandarin Chinese